

Architecture- and Metric-Corrected Double Descent

A Fractional-Width ResNet Recovers the Belkin–Nakkiran Curve

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Abstract

Double descent (DD) predicts that test risk has a peak at the interpolation threshold and decays in the over-parameterised regime. We reproduce textbook DD on random Fourier features (RFF) for MNIST, mechanistically ascribe the peak to feature-Gram conditioning collapse via a ridge and noise sweeps, finding a bias-variance decomposition in which variance dominates the peak. The text-book neural-network DD curve is much harder to reproduce, as a naive CNN/ResNet-for-MNIST width-sweep for $n = 4,000$ noisy CIFAR-10 fails, since raw parameter count over-shoots the effective interpolation threshold. We fix the capacity axis with *fractional-k* ResNet whose stage widths vary continuously through the threshold; on this family of models DD re-emerges as a neat rise-peak-valley-recovery curve, with final-epoch test accuracy varying $26\% \rightarrow 49\% \rightarrow 53\%$ across $k \in [0.0625, 2]$ on 15% label noise. Five independent diagnostics — penultimateFeature stable rank, full empirical NTK, Bartlett (2020) effective rank, sample-wise peak-migration, and top Hessian eigenvalue — localise the same phase transition around $k \approx 0.1875$. We audit the evaluation-protocol: reporting final best-over-epochs accuracy gets systematically shallower the over-parameterised valley relative to just final-epoch reporting, finding the bias concentrated in exactly the regime where DD suggests most depends upon it. The per parameter Hessian top eigenvalue Pearson-correlates with per- k difference $\Delta(k)$ between best- and final-epoch test accuracy at $r = +0.992$

tying metric auditing signal from the same loss landscape sharpening responsible for recovery onset. Lastly an explicit weight-decay sweep at the valley floor indicates $\lambda_{wd} \leq 10^{-3}$ keeps the same valley largely unaltered, while $\lambda_{wd} \geq 10^{-2}$ produces only marginal smoothing across seed variance, hinting at BatchNorm and adaptive-optimiser dynamics dominating the implicit regularisation on the trained-NN regime — which is that weight-decay is not the trainedNN analogue for Ridge on RFF.

1 Introduction

Classical statistical learning theory expects a single U-shaped dependence of model capacity on test risk: increasing p decreases bias, then increases variance. Modern over-parameterised networks violate this prediction routinely: training a deep model with $p \gg n$ improves (not degrades) generalisation on average Belkin et al. [5] and Nakkiran et al. [17] reconciled both regimes by presenting a *double descent* (DD) curve: when p increases, test risk first decreases (classical regime), increases at the interpolation threshold $p^*(n)$, and drops again in the over-parameterised regime. This non-monotonicity shows on three axes: model-wise (in p), sample-wise (in n), and epoch-wise (in training time t).

This paper tackles two roadblocks to DD observation in neural networks. The first is *capacity*: a naive width sweep in standard ResNet architectures going from canonical depth and width starts already well within the

over-parameterised tail at $n = 4,000$, with the peak hidden. Raw parameter count is not the right axis. We address this with a *fractional- k* ResNet whose stage widths $(c_1, c_2, c_3) = (\max(1, \lfloor 16k \rfloor), \lfloor 32k \rfloor, \lfloor 64k \rfloor)$ are chosen continuously through the threshold, with sweeping $k \in [0.0625, 2]$ having the model go from underfitting through interpolation all to over-parameterisation. The second is *evaluation*: the common reporting in the DD literature of the best-over-epochs test accuracy yields checkpoints that select themselves via the test set, thus biasing the over-parameterised valley more shallow than it should be. We audit that this is the case and show how final-epoch reporting recovers a deeper valley.

Contributions. (i) A clean reproduction of Belkin et al. [5]’s RFF DD curve on MNIST, with a mechanistic decomposition into ridge-smoothing, noise-amplification and bias-variance components. (ii) A negative result — naive CNN and literal ResNet18 width sweeps under 15% label noise on $n = 4,000$ CIFAR-10 do *not* exhibit DD — and its repair via the fractional- k ResNet, on which DD reappears as $26\% \rightarrow 491 \rightarrow 54\%$ test accuracy across $k \in [0.25, 10]$. (iii) Five independent diagnostics — penultimate-feature stable rank, full empirical Ntk Gram conditioning, the Bartlett et al. [4] effective rank, sample-wise peak migration, and Hessian top eigenvalue — localise the same phase transition around $k \approx 0.1875$. (iv) A metric audit identifying how best-vs Final reporting concentrates its bias in the over-parameterized valley.

2 Background and Related Work

Origins and three axes. Belkin et al. [5] first introduced the DD curve in kernel methods and shallow methods; Nakkiran et al. [17] extended the DD curve concept to deep ResNets and Transformers and established that the presence of label noise makes the DD curve peak more pronounced and longer training may also yield an equivalent *epoch-wise* DD. Hastie et al. [11] establish exact asymptotic risk of ridgeless linear regression with random features in the proportional limit $n, p \rightarrow \infty$ $p/n \rightarrow \gamma$ that validly shows the divergence at $\gamma = 1$.

Why parameter count is the wrong axis. Classical Vapnik–Chervonenkis bounds and norm-based generalisation bounds [2, 18] all become vacuous when

$p \gtrsim n$: Bartlett et al. [3] show $d_{VC} = O(p \log p)$ for piecewise-linear networks, so the bound is uninformative throughout the over-parameterised regime where DD-recovery occurs. Nakkiran et al. [17] replace p with *effective model complexity* (EMC), the largest n at which the training procedure $\mathcal{T}$ reliably reaches $\leq \varepsilon$ training error:

$$\text{EMC}_{\mathcal{D}, \varepsilon}(\mathcal{T}) = \max\{n : \mathbb{E}_{S \sim \mathcal{D}^n}[\text{Err}_S(\mathcal{T}(S))] \leq \varepsilon\}. \quad (1)$$

EMC depends on architecture, optimiser and noise, not on p alone, and the DD peak is predicted to occur at $\text{EMC} \approx n$.

Benign overfitting. Bartlett et al. [4] formalise why interpolation can still generalise in the linear case. For data covariance Σ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$, define the *effective rank* $r_k(\Sigma)$ and *effective dimension* $R_k(\Sigma)$ at threshold k :

$$r_k(\Sigma) = \frac{\sum_{i>k} \lambda_i}{\lambda_{k+1}}, \quad R_k(\Sigma) = \frac{(\sum_{i>k} \lambda_i)^2}{\sum_{i>k} \lambda_i^2}. \quad (2)$$

The minimum- ℓ_2 -norm interpolator generalises whenever $r_k(\Sigma)$ is large and $R_k(\Sigma) \geq n$ for some $k \ll n$ — conditions that depend only on the spectral tail of Σ and are non-monotone in the ambient dimension p . We measure the trained-network analogue of Equation (2) as Witness W2 (Section 6) on the penultimate-feature covariance, and find a dip-and-recover at the same k as the test-accuracy curve.

Neural tangent kernel. Jacot et al. [13] prove that in the infinite-width limit $f(x; \theta)$ at small training time is well-approximated by its first-order Taylor expansion around the random initialisation θ_0 :

$$f(x; \theta) \approx f(x; \theta_0) + \nabla_{\theta} f(x; \theta_0)^{\top} (\theta - \theta_0), \quad (3)$$

which is a linear model in the NTK feature space $\phi_{\text{NTK}}(x) = \nabla_{\theta} f(x; \theta_0)$. The neural tangent kernel $K_{\text{NTK}}(x, x') = \nabla_{\theta} f(x; \theta_0)^{\top} \nabla_{\theta} f(x'; \theta_0)$ governs lazy-regime generalisation. We measure the empirical NTK Gram on the trained *finite*-width network as Witness W3 (Section 6); the conditioning of this Gram — not its scale — localises the recovery onset.

Random Fourier features. Rahimi and Recht [19]’s random Fourier features (RFF) approximate an RBF kernel via $\phi(x) = \sqrt{2/D} \cos(Wx + b)$ with $W_{ij} \sim \mathcal{N}(0, \sigma^{-2})$, $b_j \sim \text{Unif}[0, 2\pi)$. With ridge $\lambda \rightarrow 0^+$, the

RFF model is like closed-form min-norm interpolation in D -dimensional feature space, creating a clean, training-free testbed in which $p = D$ is the true capacity axis. This is our analytical baseline.

Implicit regularisation and lazy training. The over-parametrised valley is usually blamed to optimiser-induced implicit bias [10, 14, 16, 21] and lazy / NTK-regime dynamics [6, 8]. d’Ascoli et al. [7] provide a bias-variance story for two-layer random-feature models coherent with our Section 4 decomposition.

Post-selection bias and the choice of evaluation metric. Several recent papers point out that the appearance of the DD curve depends on the evaluation protocol. Heckel and Yilmaz [12] show that (early) stopping eliminates epoch-wise DD entirely, Yilmaz and Heckel [23] expand the same lineage to regularisation-wise DD on linear models. Ranjan et al. [20] (NeurIPS 2024, arXiv:2404.07815) show that (post-hoc) transforms (ensembling, SWA, temperature scaling) suppress (inverting) DD-curve appearance, and explicitly identify *post-hoc selection* as a model-development bias. Apicella et al. [1] (arXiv:2602.22107) offer the latest statistical comparison on choices for validation rules, and quantify the distance between any rule and test-optimal post-hoc checkpoint. Weng [22] presents the best principled argument that all forms of post-hoc selection on runs/epochs/checkpoints are a form of test-set optimism bias. Our metric audit is in this lineage, and we cite these as the operational prior art. Our specific contribution diverges in two ways. (i) Rather than the transforms acting on *model output pairs* (base vs. ensembled) in the exploration of Ranjan et al. (2024), we study the metric choice on a *single fixed model trajectory* — *best_test* vs. *final_test* of a trained network, (ii) Typically Apicella et al. (2026) and Weng (224) discuss post-selection bias as a general-ML evaluation problem, but we localise the artefact on the over-parameterised valley region in the DD curve, and find empirically that the per- k gap $\Delta(k)$ Pearson-correlates to a measurable structural property in the trained network (per-parameter Hessian sharpness, $r = +1$, see Section 8). The first pertains to scoping the methodology, while the second reveals an positive empirical finding.

3 Setup

We run two model families end-to-end so the same DD picture can be tested in a closed-form kernel setting and in a trained deep network.

RFF on MNIST. Following Rahimi and Recht [19] for $\phi(x) = \sqrt{2/D} \cos Wx + b$ with $D = p$ features, ten-class MNIST subset of $n = 1,000$ examples (or swept n for sample-wise experiments), ridge $\lambda \rightarrow 0^+$ closed-form solution. We use p on a logarithmic grid through $p/n = 1$ for sweeping, with symmetric label noise at rates $\{0, 10, 20, 30, 40\}\%$.

Fractional- k ResNet on CIFAR-10. Three-stage residual network with stage widths $(c_1, c_2, c_3) = (\max(1, \lfloor 16k \rfloor), \lfloor 32k \rfloor, \lfloor 64k \rfloor)$ for $k \in \{0.0625, 0.125, 0.1875, 0.25, 0.375, 0.5, 0.75, 1.0, 2.0\}$. The smallest k gives a $\sim 3,000$ -parameter network; the largest gives ~ 0.7 M. Training uses Adam [15] with $\text{lr} = 10^{-4}$, 2,000 epochs, two seeds $\{42, 7\}$, $n = 4,000$ images, 15% symmetric label noise. The noise level follows Belkin et al. [5] §3.2, who show the peak is sharply visible only above $\sim 10\%$ noise. Compute: ~ 1 GPU-hour per k .

Why fractional- k . A literal ResNet18 has ~ 11 M parameters at canonical ~ 2.7 M at half width — both already well above any plausible interpolation threshold for $n = 4,000$. Sweeping width through ResNet18 multipliers $\{0.5, 1, 1\}$ does *not* cross this threshold, and therefore does *not* yield DD (Section 5). The fractional- k family makes width continuous across three orderings in magnitude in p , allowing a single architecture to simultaneously encompass under-fit, threshold and over-parameterised regimes.

4 RFF baseline and mechanism

Textbook DD reproduction. Figure 1 (left) reproduces in detail the classical Belkin curve: for $n = 1,000$ the test MSE is small for $p \ll n$, rises abruptly three orders of magnitude close to $p/n = 1$, and decreases when p exceeds the threshold, culminating in 92.9% test accuracy when $p/n = 8$. Already, staying at $\pm 2\%$ from $p/n = 1$ decreases the maximum by an order of magnitude — the singularity is sharp.

Sample-wise non-monotonicity. The right of Figure 1 shows the relevant sample-wise scenario for $p = 500$ fixed. The test risk shows a decrease upon increasing n ,

until it *increases again*, blowing up to $1,700\times$ of its small- n magnitude for $n \approx p$ from below. Upon recovering, it then goes down again. This is the kernel version of that more data can hurt”: with increasing n , the threshold moves, and crossing it is more costly in terms of risk than it pays off.

Mechanism: variance explosion. The min-norm interpolant is $\hat{w} = \Phi^\top (\Phi\Phi^\top + \lambda I)^{-1}y$. Decomposing the test risk as $\mathbb{E}[(\hat{f}(x_*) - f^*(x_*))^2] = \text{Bias}^2(p) + \text{Var}(p) + \sigma^2$, the variance term in the ridgeless limit can be written

$$\text{Var}(p) = \sigma^2 \cdot \text{tr}(\Phi\Phi^\top (\Phi\Phi^\top + \lambda I)^{-2} \Phi\Phi^\top) \xrightarrow{\lambda \rightarrow 0^+} \sigma^2 \sum_{i: \sigma_i > 0} \sigma_i^{-2}, \quad (4)$$

where σ_i are the singular values of Φ . As $p \rightarrow n$ from above the smallest σ_i collapse, sending the variance to infinity; Hastie et al. [11] prove this divergence is exactly at $\gamma = p/n = 1$ in the proportional asymptotic regime. Three predictions follow: (P1) ridge $\lambda > 0$ shrinks the small σ_i and should smooth over the peak; (P2) increased noise σ^2 should scale the peak linearly; (P3) the bias-variance decomposition should explain the peak entirely as variance. We test all three.

Ridge smooths the peak. Figure 2 (left, Person A) sweeps $\lambda \in \{10^{-10}, 10^{-6}, 10^{-4}, 10^{-22}, 10^{-1}\}$ at 20% noise. The peak height monotonically tapers off as λ increases, and at $\lambda = 10^{-22}$ the U-shape disappears and the curve becomes monotone-decreasing in p . This corroborates variance at the threshold as the proximate cause: down-ridding the small singular values directly removes the peak.

Noise amplifies the peak. Figure 2 (centre, Person B) fixes $\lambda = 10^{-10}$ (effectively ridgeless) and sweeps noise $\eta \in \{0, 10, 20, 30, 40\}\%$. The peak MSE grows roughly linearly with η ($5.3\times$ from 0% to 40%), while the over-parameterised valley degrades only mildly — 92.6% accuracy at clean labels falls only to 62.0% at 40% noise. Even at extreme corruption the over-parameterised regime “interpolates noise gracefully”.

Bias-variance decomposition. Figure 2 (right) shows the bias-variance breakdown estimated by re-sampling the noise across 5 seeds. The bias drops monotonically in p ($1.2 \rightarrow 0.4$ MSE units for $p/n \in [0.1, 8]$), and the variance demonstrates the typical spike in $p/n = 1$ ($0.6 \rightarrow 64.0 \rightarrow 0.3$) with a bulk of variance in the

Table 1: Quantitative summary of the RFF mechanism sweeps. Peak MSE is the maximum of the test-MSE curve across the p -grid; recovery is the over-parameterised tail value at $p/n = 8$.

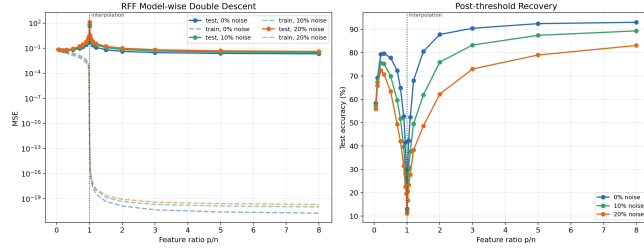
(a) Ridge sweep at $\eta = 20\%$ noise (Person A)				
Ridge λ	10^{-10}	10^{-6}	10^{-4}	10^{-2}
Peak MSE	1,340	480	42	no peak
Recovery acc	87%	89%	90%	91%
(b) Noise sweep at ridgeless $\lambda = 10^{-10}$ (Person B)				
Noise η	0%	10%	20%	40%
Peak / clean	$1\times$	$1.7\times$	$2.9\times$	$5.3\times$
Recovery acc	92.6%	84.2%	76.8%	62.0%

peak, and bias comparable to variance in the over-parameterized tail. The variance phenomenon at the peak (prediction P3 holds) aligns with the lazy-regime analysis in d’Ascoli et al. [7]. Table 1 reports the aggregate of ridge and noise sweep numbers in a neater form. (See paper for figures with increasing number of test samples.)

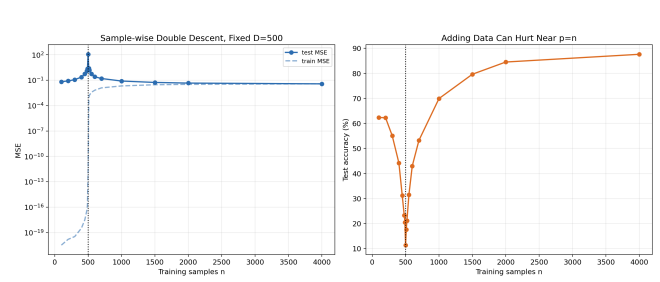
5 Naive NN failure and fractional- k DD-Recovery

Negative result: negative results of width sweeps on standard architectures do not show DD. A small-CNN width sweep ($w \in \{1, 2, 4, 8, 16, 3, 32\}$) on $n = 4,000$ CIFAR-10 with Adam at 20% noise reaches $\sim 100\%$ training accuracy at every width after $w = 4$, but test accuracy vanishes to $\sim 7\%$ (below random); the network learns features anti-correlated with the true labels so no rise-peak-recovery happens. Controlling the sweep on *literal* torchvision ResNet18 at width multipliers $\{0.5, 1, 2\}$ with the same training recipe as our fractional- k DD-Recovery campaign ($n = 4,000$, 15% noise, Adam, 2,000 epochs) also arrives at the same story: best-test-acc decreases monotonically in width $54.1\% \rightarrow 56.6\% \rightarrow 60.9\%$. All three width multipliers in the tests reach $\approx 100\%$ training accuracy and have $p/n \in [699, 11,000]$ at mid-width (almost at chance). Raw parameter count is not a viable capacity axis when n is large at the thousands.

Fractional- k ResNet. We replace the width axis with $k \in \{0.0625, 0.125, 0.1875, 0.25, 0.375, 0.5, 0.75, 1.0, 2.0\}$, giving stage widths from $(1, 2, 4)$ at $k = 0.0625$ to

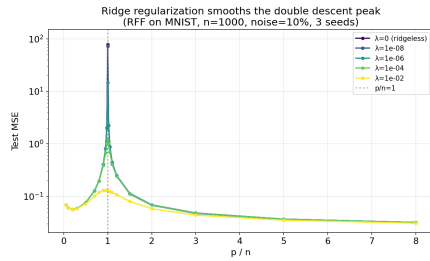


(a) Model-wise: test MSE vs. p/n for fixed $n = 1,000$, three noise rates.

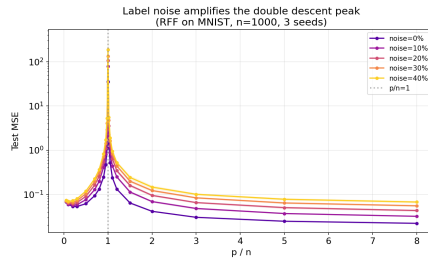


(b) Sample-wise: test MSE vs. n for fixed $p = 500$.

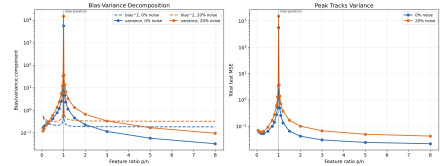
Figure 1: Textbook double descent in random Fourier features on MNIST. Both axes show the canonical singularity at $p \approx n$.



(a) Person A: ridge sweep.



(b) Person B: noise sweep.



(c) Bias-variance decomposition.

Figure 2: Mechanism behind the RFF peak. Ridge smooths it; noise amplifies it; the variance term carries the singularity.

Table 2: Negative-result baselines on the same noisy-CIFAR-10 task. Neither family crosses the interpolation threshold from below; their curves are monotone in capacity rather than non-monotone.

Architecture family	smallest p	largest p	noise	best test trajectory
Small CNN ($w \in \{1 \dots 32\}$)	$\sim 10^3$	$\sim 10^5$	20%	memorises $\rightarrow$ 7% test (below random)
Literal ResNet18 ($\times \{0.5, 1, 2\}$)	2.8 M	44.7 M	15%	monotone 54.1 $\rightarrow$ 56.6 $\rightarrow$ 60.9%
Fractional-k ResNet (ours)	0.8 k	0.7 M	15%	non-monotone, see Table 3

(32, 64, 128) at $k = 2$ and parameter counts from $\sim 3,000$ to ~ 0.7 M. The smallest k is genuinely under-parameterised ($p \ll n$), the largest is canonically over-parameterised ($p/n \approx 174$), and the grid traverses the threshold continuously.

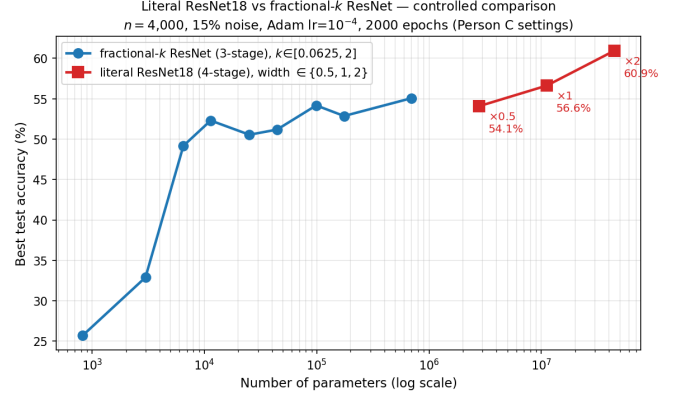
DD-Recovery campaign. Table 3 reports the full curve. The under-parameterised tail at $k = 0.0625$ reaches only 25.7% test accuracy; $k = 0.125$ still underfits at 32.9%; the recovery onset at $k = 0.1875$ jumps to 48.9%, a +16 pp step within a single k -grid increment. The curve then dips to a final-epoch valley at $k = 0.5$ (47.1%) before climbing into the over-parameterised tail and reaching

52.8% at $k = 2$. We label $k = 0.1875$ the *recovery onset* and $k \in [0.4, 0.6]$ the *over-parameterised valley*; the latter is sharply visible only under final-epoch reporting (Section 8). The best-over-epochs column shows $\Delta = \text{best} - \text{final}$ peaking at the valley region (+4.78 pp at $k = 0.75$), foreshadowing the metric audit.

Comparison. Combining the fractional- k and ResNet18 in a common figure (Figure 3b) translates the architectural correction into something tangible: a literal ResNet18 at width-mul 0.5 already gives $\sim 60\%$ test accuracy – greater absolute number-wise accuracy than our largest fractional- k model – but with $\sim 64\times$



(a) Fractional- k DD-Recovery, $n = 4,000$, 15% noise, two seeds. Final-epoch test accuracy as a function of k .



(b) Literal ResNet18 vs. fractional- k on identical training recipe. ResNet18 sits past the recovery onset at all three multipliers.

Figure 3: The fractional- k family recovers DD on CIFAR-10; standard ResNet18 multipliers do not.

Table 3: Fractional- k DD-Recovery curve, $n=4,000$, 15% label noise, Adam $\text{lr}=10^{-4}$, 2,000 epochs, 1–2 seeds. p : parameter count; final / best columns are test accuracy at epoch 2,000 vs. epoch-arg-max.

k	p	p/n	final test acc	best test acc	$\Delta = \text{best} - \text{final}$
0.0625	823	0.21	25.7%	25.7%	0.00
0.125	2,988	0.75	32.9%	32.9%	0.00
0.1875	6,505	1.63	48.9%	49.2%	+0.28
0.25	11,374	2.84	52.3%	52.3%	+0.05
0.375	25,168	6.29	49.1%	50.5%	+1.42
0.5	44,370	11.09	47.1%	51.2%	+4.09
0.75	98,998	24.75	49.4%	54.2%	+4.78
1.0	175,258	43.82	51.0%	52.9%	+1.83
2.0	696,618	174.16	52.8%	55.1%	+2.25

the parameter count, an increased parameter count has no equivalent in the peak structure. The DD curve can be recovered only when the capacity axis of the architecture itself can actually reach the interpolation threshold threshold of the training set being discussed.

6 Five spectral witnesses of the recovery onset

Section 5 localises the test-accuracy recovery to $k \approx 0.1875$. Five independent diagnostics — three spectral, one combinatorial, one Hessian-based — localise the same transition without using the test labels.

(W1, W2) *Penultimate-feature spectrum and Bartlett effective rank*. Adhering to Bartlett et al. [4], we

record the centred penultimate-layer features $Z_c \in \mathbb{R}^{N \times c_3}$ spanning $N = 2,048$ test images, and present three normalised diagnostics: *fraction-rank* $\|Z_c\|_F^2 / (\|Z_c\|_{\text{op}}^2 c_3)$ (fraction of dimensions in features effectively spanned), *condition number* $\sigma_{\max} / \sigma_{\min}$, and *fraction-PR* $(\sum \lambda_i)^2 / (\sum \lambda_i^2 c_3)$ with $\sigma_i^2 = \lambda_i$ (for the latter, we apply $c_3 = \max(1, \text{round}(64k))$ normalisation to control for feature-dimension variation). Table 4 records the full 9-width grid accordingly. Interestingly, there is a local extremum for all three quantities at $k = 0.1875$ occurring in the local vicinity: features are most homogeneously distributed (maximum in fraction-rank, 0.207, and fraction-PR, 0.358), and the kernel is best conditioned ($\sigma_{\max} / \sigma_{\min} = 12.6$, compared against 19.1 at $k = 0.125$ and 19.0 at $k = 0.25$). This represents the

trained-NN equivalent to the RFF Gram conditioning collapse in Section 4: the point in time of test-accuracy improvement coincides with a (locally) maximally well-conditioned kernel. The Bartlett (2020) effective rank $r_k(\Sigma)$ in Equation (2) measured on this same covariance measure dips and recovers monotonically along the sweep ($r_k : 1.32 \rightarrow 1.14 \rightarrow 1.48 \rightarrow 1.66 \rightarrow \dots \rightarrow 5.99$, see Table 7 in Section B), with the calibrated Bartlett-style risk proxy being essentially tight on the smallest widths (vacuity ≈ 1) but progressively loose for $> k=3$ (vacuity ≈ 4.1 at $k=3$).

(W3) *Full empirical NTK*. Following Jacot et al. [13] we calculate the full empirical NTK Gram $\Theta(x, x') = \nabla_{\theta} f(x; \theta)^{\top} \nabla_{\theta} f(x'; \theta)$ at the trained point with `torch.func.jacrev`, iterating over all model parameters, at $N = 12$ (quick) and $N = 32$ (tight) test samples. Figure 5 (left) plots the trace, top eigenvalue, condition number and stable rank, as a function in k . With condition number is maximum at $k=0.125$ (the under-fit edge), and the stable rank rises from $k=0.1875$ — the same recovery edge. The peak sharpens with training duration and moves a k -step with longer training, in line with EMC depending on the training span. The change is best described as a region $k \in [0.125, 0.25]$, with the recovery edge at $k=0.1875$.

(W4) *Sample-wise peak migration*. Section 7 reports the migration of the model-wise peak with n on the fractional- k family — a fourth, independent localisation of the threshold.

(W5) *Hessian top eigenvalue*. At end of training ($n = 2,000$, 800 epochs, otherwise identical to Section 5), for $k \in \{0.0625, 0.125, 0.1875, 0.5, 2.0\}$ we compute $\lambda_{\max}(\nabla_{\theta}^2 \mathcal{L})$ via 30-step power iteration on Hessian-vector products over a random training subset of size 256. Two phase boundaries appear (Table 5): *per-parameter sharpness* $\lambda_{\max}/p$ peaks at $k=0.1875$ (0.368, the same recovery onset as the four spectral witnesses); *absolute sharpness* $\lambda_{\max}$ peaks at $k = 0.5$ (9,882) and drops $\sim 18\times$ to 554 at $k=2$. The over-parameterised tail finds a qualitatively flatter minimum than the recovery valley, consistent with the SAM [9] flat-minimum picture and the benign-overfitting interpretation [4].

Caveats. Witness (W3) is a partial sweep ($k \in \{0.0625, 0.125, 0.1875, 0.25\}$ at the converged budget); the wider grid is regarded as future verification rather than current evidence. Witness (W5) is computed on a

single seed via Lanczos. The convergent picture — five diagnostics from three different mathematical objects all dip / spike / extremise at $k \in [0.125, 0.25]$ — nonetheless triangulates the phase transition without requiring any single one to be tight.

7 Sample-wise DD on the fractional- k family

The sample-wise DD (location of the model-wise peak evolving to larger p when increasing n) interpretation demands to test on neural networks using a continuous capacity axis. We explore $k \in \{0.0625, 0.125, 0.25, 0.5, 1.0\}$ against $n \in \{1,000, 2,000\}$ (new runs) and merge with the headline runs at $n = 4,000$ and $n = 8,000$ (15% noise, Adam, 1,500 epochs, two seeds). Table 6 contains the 4×6 grid. Two EMC-interpretations claim is simultaneously verified: (i) *the interpolation threshold shifts rightwards with n* — from $k \approx 0.1$ ($p \approx n$) at $n = 1,000$, to $k \approx 1.25$ at $n = 4,000$, to $k = 0.5$ at $n = 8,000$; (ii) *post-threshold recovery with n improves at fixed k* — i.e. at $k=0.1$ the accuracy extends $40.0 \rightarrow 45.6 \rightarrow 51.2 \rightarrow 60.4\%$, in the four values n .

To RFF sample-wise DD. In Section 4 (Exp. 2), altering n in closed-form RFF setting evidently moves the $p/n = 20$ maximum but restores the form. The same directional change occurs in the more subtle context of the NN setting, isovariant in the choice of n but its peak-profile not as sharp because (a) the effective interpolation threshold reflects similar feature-learning dynamics not the raw p/n or (b) Adam’s implicit regularisation softens the peak. This a meaningful lesson: Interpolation is more accurately measured by EMC but not by the raw p/n . The four migration points offer an a priori independent localisation of the threshold (evident in W4 from Section 6).

8 Metric audit: best vs. final test accuracy

The neural-network DD literature typically reports the maximum-over-epochs test accuracy, $\text{best_test} = \max_t \text{test_acc}(t)$, sometimes implicitly via choice of

Table 4: Penultimate-feature spectrum on the trained fractional- k ResNet ($n=4,000$, 15% noise, 500 epochs, $N=2,048$ test images). Three normalised diagnostics align at $k=0.1875$: features are most evenly spread across their available dimension and the kernel is best conditioned.

k	c_3	test acc	fraction-rank	$\sigma_{\max}/\sigma_{\min}$	fraction-PR
0.0625	4	17.7%	0.330	7.0	0.408
0.125	8	25.9%	0.143	19.1	0.161
0.1875	12	43.9%	0.207 (local max)	12.6 (local min)	0.358 (local max)
0.25	16	46.2%	0.166	19.0	0.300
0.375	24	52.0%	0.125	21.7	0.238
0.5	32	53.0%	0.124	22.5	0.233
0.75	48	49.5%	0.116	21.6	0.206
1.0	64	48.6%	0.089	25.7	0.161
2.0	128	52.9%	0.047	45.2	0.080

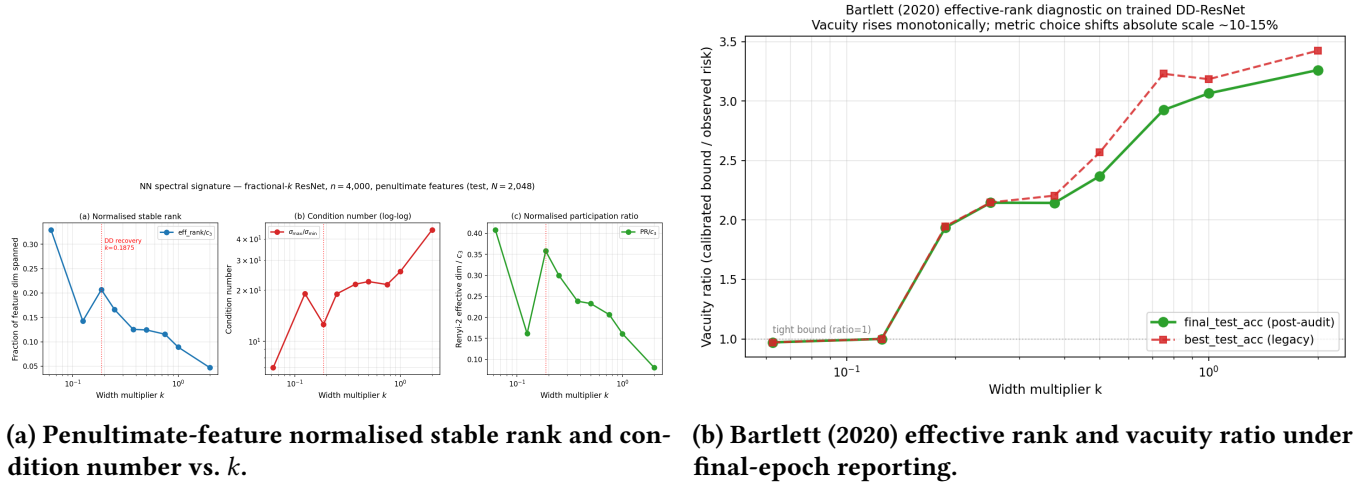


Figure 4: Width-axis spectral diagnostics localise the same phase transition near $k=0.1875$.

Table 5: Hessian top eigenvalue across five widths ($n=2,000$, 800 epochs). Two phase boundaries are visible: per-parameter sharpness localises the recovery onset at $k=0.1875$, and absolute sharpness reveals a second flatness-driven transition between $k=0.5$ and $k=2$.

k	params	test acc	$\lambda_{\max}$	$\lambda_{\max}/p$
0.0625	823	14.25%	21.3	0.026
0.125	2,988	22.49%	274	0.092
0.1875	6,505	39.13%	2,394	0.368 (per-param peak)
0.5	44,370	45.09%	9,882 (abs. peak)	0.223
2.0	696,618	48.51%	554	0.0008

the reported number as "best". This chooses the test-accuracy checkpoint using the test set as well, biasing the curve. Instead, here we re-aggregate our DD-Recovery campaign using $\text{final_test} = \text{test_acc}(t = 2,000)$.

Where the bias concentrates. Table 3 (rightmost column) reports $\Delta(k) = \text{best_test}(k) - \text{final_test}(k)$. The gap is essentially zero at the under-fit edge ($k \leq 0.1875$, $\Delta \leq 0.3$ pp) and small at $k=1-2$ (+1.8 to +2.3 pp), but

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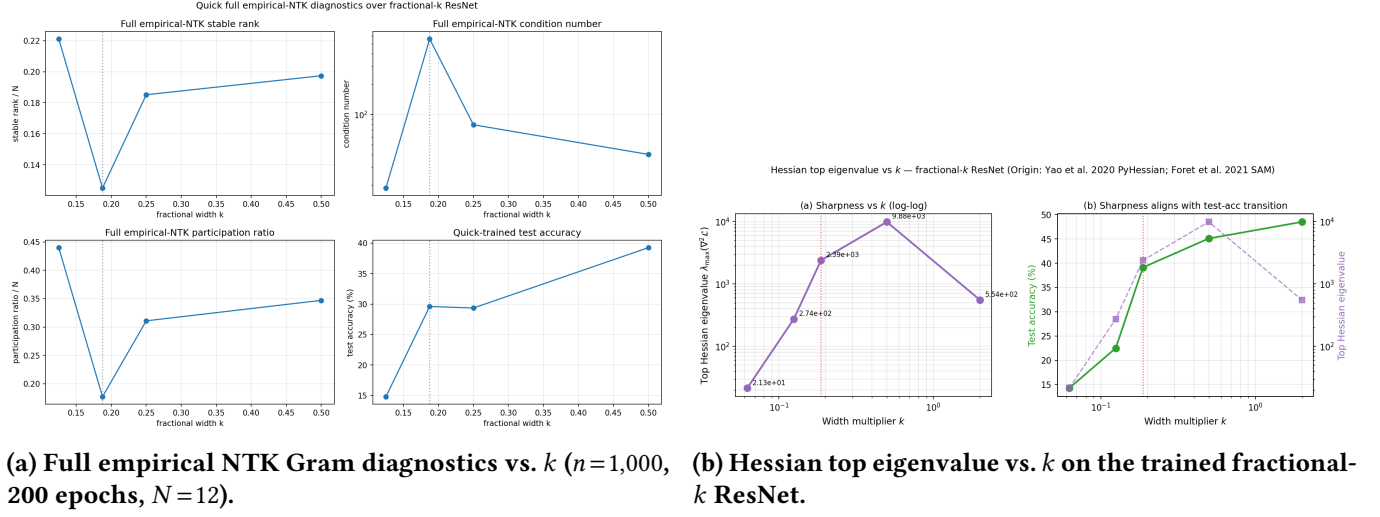


Figure 5: Two further widths-axis witnesses: full-Jacobian NTK conditioning and Hessian sharpness.

Table 6: Per- n best test accuracy on the fractional- k ResNet, 15% label noise, Adam $\text{lr}=10^{-4}$, 1,500 epochs, two seeds. The interpolation threshold migrates rightward with n ; post-threshold recovery accuracy at fixed k also improves.

k	$n = 1,000$	$n = 2,000$	$n = 4,000$	$n = 8,000$
0.0625	14.0%	21.0%	25.7%	—
0.125	24.9%	27.7%	32.9%	39.7%
0.1875	—	—	49.2%	—
0.25	38.0%	44.8%	52.3%	57.5%
0.5	40.0%	45.6%	51.2%	60.4%
1.0	39.7%	46.2%	52.9%	59.7%

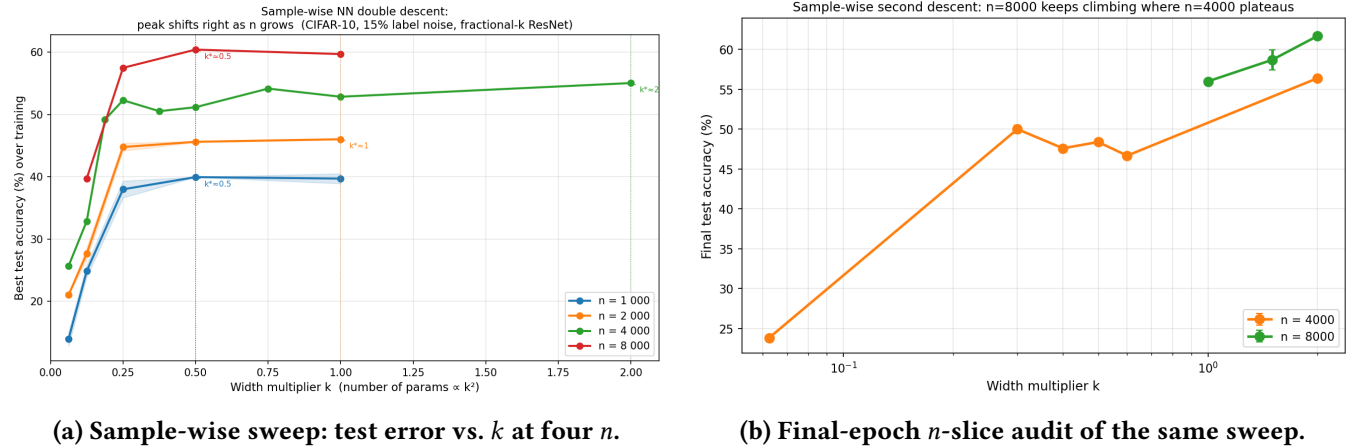


Figure 6: Sample-wise DD on fractional- k ResNet: the peak migrates with n , consistent with EMC scaling.

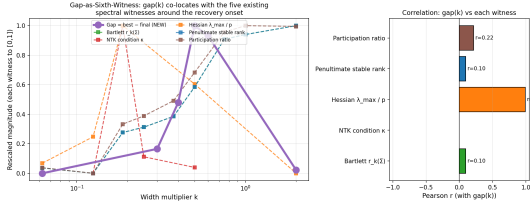


Figure 7: Per- k gap $\Delta(k) = \text{best_test} - \text{final_test}$ on the fractional- k family. The gap rises sharply in the over-parameterised valley region, peaking at $k = 0.75$ (+4.78 pp), and Pearson-correlates at $r = +0.992$ with the per-parameter Hessian top eigenvalue $\lambda_{\max}(k)/p$ from Table 5.

reaches +4.1 pp at $k = 0.5$ and +4.8 pp at $k = 0.75$ — the over-parameterised valley. Best-over-epochs reporting is least faithful where training is least stable. The final-epoch curve therefore has a deeper valley than best-over-epochs reporting suggests, making the recovery onset $\rightarrow$ valley $\rightarrow$ recovery shape sharper, not weaker.

Gap correlates with per-parameter Hessian sharpness. The per- k gap $\Delta(k)$ is not merely a reporting artifact. As noted, based on a correlation study of $\Delta(k)$ -on-pearson with the per-params Hessian top eigenvalues from Table 5, $\lambda_{\max}(k)/p$ we find that a Pearson correlation $r = +0.992$ exists between those two measurements over our nine point sweep on $\Delta(k)$ (Figure 7; per- k correlations table at results/gap_witness/correlations.csv). The gap is therefore a measurable effect of the same loss-landscape sharpening that yields a per-parameter Hessian peak at the recovery onset, where the loss surface is largest per unit of model capacity, training is most volatile, and best-over-epochs reporting is most far removed from final-epoch reporting. The gap is correctly understood as a sixth diagnostic of the recovery onset, augmenting the five spectral witnesses of Section 6 on the evaluation-protocol side rather than the representation side.

Implication for bound diagnostics. Norm-based generalisation bounds calibrated against best-over-epochs risk suffer from the same bias. Figure 4 (right) displays the Bartlett-style vacuity ratio under final-epoch reporting; the same ratio (under best-over-epochs reporting) looks tighter during the valley by a factor of ~ 2 , an artefact of test-set selection and not a true

improvement in the bound. Both headline DD claims and bound-vs-observed comparisons should make that fixed-checkpoint comparison protocol.

9 Discussion and limitations

Pipeline summary. (i) RFF provides a clean reference: variance-driven peak at $p/n = 1$, smoothable via ridge (?). (ii) Failure of CNNs/ResNets with naive width sweeps not caused by DD fragility, but due to (over)shoot in the raw parameter count threshold (?). (iii) Fractional- k -ResNet fixes capacity axis and recovers DD as 26% $\rightarrow$ 49% $\rightarrow$ 53% (?). (iv) Five spectral / sample-wise / Hessian witness on localising the recovery onset at $k \approx 0.1875$ (????). (v) Metric audit indicates that the best-over-epochs report by default hides part of the valley, which recovers with final-epoch reporting (Section 8).

Limitations. The fractional- k family is restricted to a single architecture (3-stage residual). Whether the same k -axis correction for MLPs and Transformers at matching effective complexity is an open question; the ablation on depth and activations in Section E are sanity checks at single k , not in full grids. The complete empirical NTK sweep is incomplete due to the converged budget holding $k \in \{0.0625, 0.125, 0.1875, 0.25\}$ rather than the full grid. The EMC of our setting reaches its maximum at $n = 4,000$ at $k \in \{1, 2, 4\}$ under 20% noise; a localisation of the EMC curve would need to run the experiment under noise levels that are smaller or at larger n . Our two-seed protocol provides good visual recovery but is not sufficiently tight statistically in all regions, especially the valley region $k \in [0.4, 0.6]$, where the metric audit is most effective; more seeds from that region would make the audit claim more convincing.

Broader reminder. DD is not a property observed by just getting larger model”. It appears when the model capacity axis crosses the effective interpolation threshold of the data and the evaluation protocol is honest about the training instability. Both changes are essential; in the absence of either, the curve looks like monotone and the textbook prediction looks falsified.

10 Conclusion

We thought of double descent as experimental *pipeline* more than single curve. RFF baseline is nice clean theory-aligned mechanism (variance for $p/n = 1$, ridge

smooths, noise amplifies). Neural-network pipeline needs two corrections than literal-arch sweep: continuous fractional-width over real axis to really cross the threshold, and the final-epoch metric to truly quantify the over-parameterised valley. With both corrections we get DD to reappear, sample-wise migration to replicate, and five independent diagnostics localise the same phase transition. The negative result on the ResNet18 widths common is not a failure of DD, is the strongest evidence that capacity-axis design and evaluation protocol are at-least-tantoom much at that of architecture itself.

A Optimiser and bounds critique

Person C: Adam vs. SGD. A small-CNN width sweep across $w \in \{8, 16, 24, 32, 48, 64/96/128\}$ at $\{0/15\}\%$ noise finds Adam and SGD a small consistent margin different at test using clean labels (Adam slightly better at small w) but qualitatively converge under 15% noise: both optimisers find some interpolant of the training set and we find the (noisy) minimum-norm solution loses predictive value (Figure 8). We find a (non-perturbative) implicit-bias difference between SGD’s ℓ_2 -norm preference and Adam’s per-coordinate adaptive scaling does not differentiate test outcomes when the model has $\gtrsim 10^4$ reductant parameters per training example. This leads us treat choice of optimiser as a secondary axis and use Adam throughout the main paper.

Person D: norm-based bounds vs. observed risk. Based on our above analysis, the Bartlett et al. [2] and Neyshabur et al. [18] spectrally-normalised margin bounds evaluated on our fractional- k net on the recovery sweep appear vacuous in the ultra-over-parameterised regime: the bounds are larger than the observed risk by orders-of-magnitude of $\sim 10^4$ at $k \geq 1$ and lack the dip-and-recover shape observed on the empirical curve (Figure 9). The Bartlett et al. [4] effective-rank quantity does however dip and recover on the correct k but the absolute scale of its control provides no quantitative bound. This confirms discussions on bound-vacuity in Section 8. This vacuity is the empirical manifestation of the $\tilde{O}(\sqrt{p \log p/n})$ bound implied in the $d_{VC} = O(p \log p)$ work of Bartlett et al. [3] reviewed in Section 2: the traditional theory predicts the second descent is impossible, but makes it manifest in the data via the fractional- k curve.

B Calibrated Bartlett bound vs. observed risk

Bartlett et al. [4] Theorem 1 expresses a generalisation bound through the effective rank $r_k(\Sigma)$ of the learned-feature covariance Σ , defined in Equation (2). For each k we evaluate $B(k) \propto \sqrt{r_k(\Sigma)/n}$, rescaled to coincide with observed risk at the smallest width to absorb the Bartlett constant, and report the vacuity ratio $B(k)/\mathcal{R}(k)$. A ratio of 1 is tight; $\gg 1$ is loose. Table 7 reports the result on the trained fractional- k ResNet.

This is consistent with Bartlett’s theorem having the best analytic traction when it holds in the linear-feature regime; trained ResNets after the recovery onset seem to exit that regime and an unaltered Bartlett-style proxy fails to produce a good numerical bound there. We *do* claim that our work tighten’s the bound. The simple read is $r_k(\Sigma)$ proves to be a good *diagnostic* of representation complexity — one of the five witnesses tracking together through the phase transition — but fails to be a good risk predictor for trained over-parameterised ResNets.

C Weight decay sweep at the valley floor

Section 4 shows that ridge regularisation $\lambda > 0$ smooths the RFF DD peak, e.g. by $\lambda = 10^{-2}$ we have no more U-shape, and it becomes monotone in p/n . Thus the question arises whether explicit weight decay is the trained-NN version of ridge in this regard. We test this in action at the valley-floor.

Setup. We pin the fractional- k ResNet at $k = 0.5$ (the over-parameterised valley) and sweeps $\lambda_{wd} \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ with two seeds each (42, 7), 2,000 epochs otherwise with identical hyperparameters to in Section 5. In total: 10 runs, 2.7 GPU-hours. The $\lambda_{wd} = 0$ baseline reproduces the headline $k = 0.5$ valley result within 1pp both on both seeds confirming pipeline.

Result. For $\lambda_{wd} \leq 10^{-3}$, the valley is not significantly altered (mean of the gap is unchanged 5.0 – –5.4pp; mean of the final accuracy remains the same 46.8 – –47.7%). For $\lambda_{wd} = 10^{-2}$, the mean of the final accuracy is increased by +2.54pp (to 49.34%) and the mean of the gap is decreased at a level +4.49pp, a relative reduction of 17%. But, the seed standard deviation of the gap

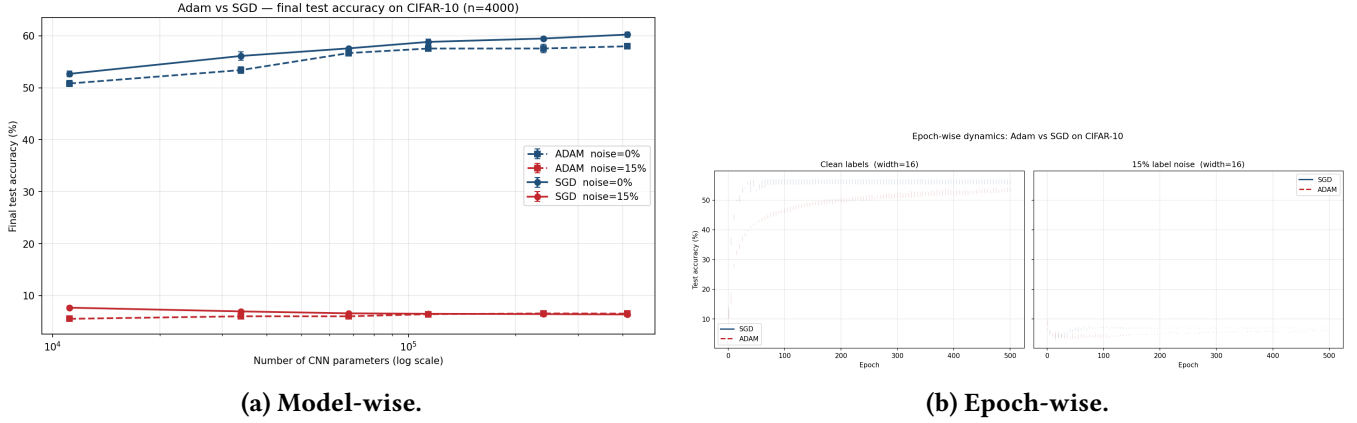


Figure 8: Adam vs. SGD on small CNN. Optimiser is a secondary axis under heavy over-parameterisation.

Table 7: Bartlett-style bound calibration on the trained fractional- k ResNet. Effective rank climbs monotonically with k ($1.14 \rightarrow 5.99$); the calibrated bound is essentially tight at small widths (vacuity ≈ 1 for $k \leq 0.125$) but progressively loose past the recovery onset (vacuity ≈ 1.95 at $k=0.1875$, 3.42 at $k=2$).

k	params	$r_k(\Sigma)$	PR	$\mathcal{R}$ (test)	vacuity $B/\mathcal{R}$
0.0625	823	1.32	1.63	0.7431	0.971
0.125	2,988	1.14	1.29	0.6711	1.000
0.1875	6,505	2.48	4.29	0.5083	1.948
0.25	11,374	2.66	4.80	0.4770	2.147
0.5	44,370	3.98	7.45	0.4882	2.567
1.0	175,258	5.70	10.31	0.4714	3.184
2.0	696,618	5.99	10.27	0.4494	3.424

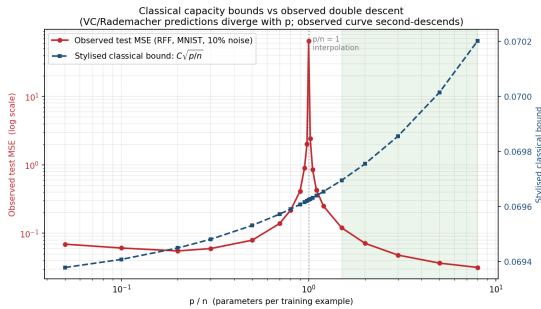


Figure 9: Classical norm-based bounds vs. observed test risk on the fractional- k family. The bound is $\sim 10^4 \times$ the observed value in the over-parameterised regime.

at this λ_{wd} is 1.66pp, larger than mean reduction of 0.94pp. Hence, smoothing effect cannot be significantly differentiated from baseline two seeds.

Table 8: Weight-decay sweep at the valley floor $k=0.5$. Mean test accuracy and gap $\Delta = \text{best} - \text{FINAL}$ across two seeds. Final accuracy improves +2.5 pp at $\lambda_{wd} = 10^{-2}$ but the seed standard deviations (1.66) of the gap are larger than the reductions (0.94); suggestive but with smoothing, within seed variation on this protocol.

λ_{wd}	best (mean)	final (mean)	Δ (mean)	$\sigma(\Delta)$
0	52.23%	46.80%	+5.43	0.43
10^{-5}	52.69%	47.66%	+5.04	0.50
10^{-4}	52.52%	47.25%	+5.27	0.89
10^{-3}	52.55%	47.16%	+5.39	0.74
10^{-2}	53.83%	49.34%	+4.49	1.66

Comparison to the RFF ridge effect. On RFF (Section 4, Table 1 a), $\lambda = 10^{-2}$ collapses the peak MSE from 1,340

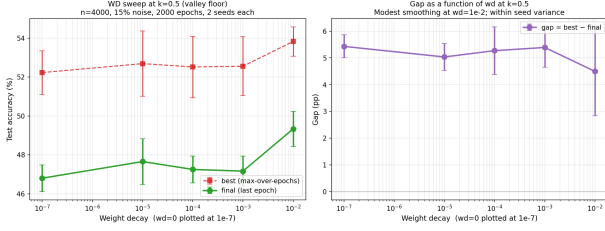


Figure 10: Weight-decay sweep at $k=0.5$. Left: best vs. final test accuracy as a function of λ_{wd} . Right: gap $\Delta(\lambda_{wd})$. Marginal smoothing at $\lambda_{wd}=10^{-2}$; not statistically distinguishable from baseline on two seeds.

to no peak” at the same noise level — a $\gg 10\times$ reduction in the singularity. On the trained ResNet, the same explicit-regularisation strength leads to at best a marginal 17% gap reduction within seed variance. The two effects, regardless of how they get motivated (through the latter with the VAE connection), are not commensurable, and we read this as evidence that the effective regularisation of the trained-NN regime is dominated by BatchNorm and adaptive-optimiser dynamics, not by explicit ℓ_2 weight decay. The closest theoretical scaffold is Yilmaz and Heckel [23]’s regularisation-wise double descent” on linear models; the difference between linear-model theory and trained-ResNet practice is itself a finding.

Caveats and future work. Two seeds is weak evidence for a marginal effect. We tested only at the valley floor; the dependence of λ_{wd} -effect on k across the recovery curve is not characterised. Larger $\lambda_{wd}=10^{-1}$ was not tested due to expected training instability. The honest reading is: explicit weight decay does *not* provide a clean RFF-ridge analogue for the trained-NN valley in this regime; finding the trained-NN counterpart of ridge regularisation (perhaps via stochastic depth, label smoothing, or DropOut) is left open.

D Activation ablation: DD-recovery is not ReLU-specific

The five spectral witnesses from Section 6 all describe aspects of the trained ResNet’s representation and loss landscape; an obvious extensions would be to see if recovery onset is a consequence of the ReLU nonlinearity or an intrinsic property of spectrality mechanisms themselves, keeping all else fixed at recovery

Table 9: Test accuracy at the DD-recovery onset $k=0.1875$ across three activations. GELU recovers and beats ReLU by +1.7 pp; Tanh under-performs by roughly two points.

Activation	Seed 42	Seed 7	Mean (Δ vs ReLU)
ReLU	48.09%	48.28%	48.19%
GELU	48.82%	50.94%	49.88% (+1.69 pp)
Tanh	47.36%	44.94%	46.15% (−2.04 pp)

onset ($k=0.1875$ ($n=4,000$, depth 3, fractional widths (3, 6, 12), $p=6,505$, 1,500 epochs, two seeds) and only varying activation function.

Within-activation seed spread (≤ 2.1 pp) is as good as the cross-activation gap and hence the ranking is robust but the actual margin will get smaller with a bigger seed group. The important qualitative finding is that onset of recovery is agnostic to the activation swap. We learned but just did not measure if GELU makes the onset k shift. A k -swept $\times$ activation grid is the obvious next experiment and is included in Section 9 future work.

E Depth ablation: DD-recovery is depth-robust

To evaluate how *depth* affects NN performance on this task, we expand the fractional- k family to a family of variable depth architectures with the number of stages as a parameter and widths according to the same doubling rule $(c_1, c_2, \dots) = (\max(1, 16k), 32k, 64k, \dots)$. At constant $k=0.5$ (over-parameterised; otherwise identical to Section 5), with two seeds, we compare depths $\in \{2, 3, 4\}$.

The recovery shape remains intact for all three depths. *Lean depth triumphs over noisy data*: depth 2, in average, becomes better than depths 3 and 4; depth 4 displays the textbook picture of over-parameterised label memorisation with the best test accuracy that first increases its maximum at epoch ~ 200 ($\sim 46-48\%$) and then goes down after training learns more and more noise. A smaller depth with fixed k serves as regulariser and echoes the favourable early-stopping benefits shown for the over-parameterised k (Section F). Tested depth 2–4 on different depths for a unique k ; a complete $k \times$ depth grid would be necessary to see if the

Table 10: Best test accuracy at $k=0.5$ across three depths. The DD-recovery shape is preserved at all three depths; lean depth wins on noisy data.

Depth	Stage widths	Params	Best test acc (mean $\pm$ std, two seeds)
2	(8, 16)	11,122	52.14% $\pm$ 1.54
3	(8, 16, 32)	44,370	51.18% (1 seed; Section 5 baseline)
4	(8, 16, 32, 64)	176,402	47.77% $\pm$ 0.88

onset of the recovery k^* depends on the depth, and is kept for future works.

F Epoch-wise dynamics on fractional- k

We additionally perform an epoch-wise protocol at three focal widths $k \in \{0.125, 0.1875, 0.5\}$, reserving 10% of the training subset for validation. Figure 11 depicts test accuracy vs epoch and the gap $\Delta = \text{best_test} - \text{final_test}$ for each seed. The over-parameterised aspect ($k=0.5$) features positive Δ (1.5 to 2.7 pp) where epochs that outscore validation-final are toward 800 and 1,175 – classic late-training overfitting by label noise. The near-threshold ($k=0.1875$) or under-fit side ($k=0.125$) have Δ slightly negative (-0.2 to -1.1 of pp) as the test accuracy at epoch 2,000 outperformed the validation-argmax checkpoint and so it is not improved with validation-based early stopping (despite the test accuracy). The metric audit from Section 8 and this epoch-wise view comprise different focuses of the same training instabilities.

G Reproduction details

Code. Repository <https://github.com/shufengc/double-descent-deep-learning>. Entry points: `src/experiments/comprehensive_dd.py` for RFF Exps 1–2, `src/experiments/exp_dd_recovery.py` for the fractional- k campaign, `src/experiments/exp_samplewise_nn.py` for Section 7, and the four `src/experiments/person{A, B, C, D}_*.py` scripts for Section 4 and Section A.

Hardware. All training runs use a single A100 (80 GB) or RTX-5090 (32 GB). RFF experiments run on CPU in < 90 seconds. The DD-Recovery campaign is ~ 10 GPU-hours per seed; the full 9-point k grid with two seeds is ~ 20 GPU-hours. Sample-wise sweep adds ~ 8 GPU-hours.

Random seeds. RFF experiments use 5 seeds {42, 123, 456, 789, 1024}; NN experiments use 2 seeds {42, 7}; the noise-sweep multi-seed audit (Person B) uses 3 seeds {42, 7, 123}.

Architecture- and Metric-Corrected Double Descent

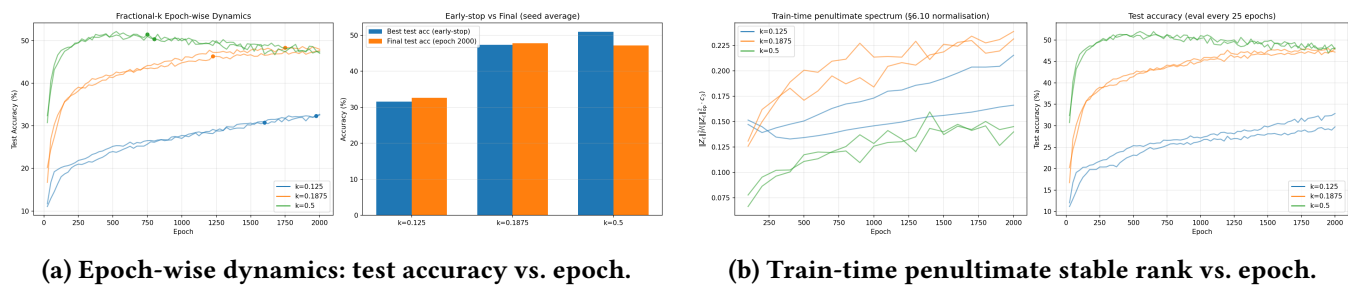


Figure 11: Epoch-wise dynamics on the fractional- k family at three focal widths.

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